

Patel Shlok Umesh

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PROFESSIONAL SUMMARY

Backend developer passionate about designing clean APIs, optimizing databases, and building robust system architectures. I enjoy solving complex structural problems, whether that means orchestrating data extraction pipelines in Python or building out scalable MVC architectures in Java. I focus on writing efficient, maintainable code and engineering high-performance systems that scale smoothly.

EDUCATION

Bachelor of Engineering in Computer Engineering

2024 – 2028 (Expected)

LJ Institute of Engineering and Technology (LJIET), Ahmedabad, Gujarat

- Focus on System Design, Data Structures, Algorithms, and Database Management.

Bachelor of Science in Data Science

2024 – 2026 (Expected)

Indian Institute of Technology Madras (IIT Madras) - Online Degree Program

- Specialization in Python Programming, Statistics, and Machine Learning.

TECHNICAL SKILLS

- Languages:** Java, Python, Rust, SQL, TypeScript
- Backend Frameworks:** FastAPI, Node.js, Maven, RESTful APIs
- Databases:** PostgreSQL, Supabase, SQLite, MySQL
- Architecture & Tools:** MVC, Agentic Graphs, OOP, Docker, Git, GitHub, JDBC

PROJECT EXPERIENCE

Aeres-IDE Local Architecture

Rust, Python, Tauri, FastAPI

- Designed core backend microservices utilizing Rust and FastAPI, efficiently handling local file processing with highly optimized I/O operations.
- Implemented Causal Maps and Temporal Lens profiling to handle autonomous codebase tracing and real-time LLM interaction under 200ms latency.

MediBot V2 Backend

FastAPI, Python, Groq LLMs

- Architected the cognitive core of a multi-agent medical triage system using FastAPI, processing real-time contextual chat for concurrent users.
- Engineered an advanced Agentic Graph that securely routes patient symptoms through specialized LLM agents with 95%+ accuracy scoring.

Intelligent Document Extraction Pipeline

Python, PyMuPDF, OCR

- Engineered a high-performance Python data extraction pipeline integrating Tesseract and EasyOCR to parse 100+ page PDFs in seconds.
- Developed automated regex-based entity extraction layers to reliably parse and output structured data (emails, phones, key-value pairs).

NeonPulse (Desktop Music Player)

Java 17, JavaFX, SQLite, Maven

- Developed an offline music player utilizing rigorous OOP concepts, executing high-speed SQL queries via JDBC for persistent playlist tracking.